

Team members:

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Title: Pakistan's Digital Payment Analytics

1-Domain and Source identification:

Project Concept:

Our project focuses on the digital payments and the banking sector in Pakistan. The aim is to see how the payment method has changed over time, particularly how is the shift from the traditional cash only payment to payment using digital methods.

We will analyse payment activities such as: ATM transactions, mobile banking, internet banking, Point of sales transactions, e-commerce payments, card payments, and other digital payment methods.

Data Source:

The main data acquisition for our project would be the official site of SBP (State Bank of Pakistan) and its official payment review system.

The link for the site would be: <https://www.sbp.org.pk/our-operations/publications/payment-systems-review/365>

The data is available in the form of reports which are published every quarter of every year so we will be basically using it.

Full Load Ingestion:

For the full load ingestion we will collect almost all the available data which will almost cover from 2006-2026 a span of about two decades. This will provide the historical baseline for our project and will allow us to study the how the trend changed and how it will/ may change in the upcoming future.

Incremental Load Ingestion:

The State Bank of Pakistan's site publishes these quarterly reports periodically. So we will collect the new data and the pipeline will check if the data is new or is it something already available in the initial load and if new pipeline will add only that data.

2- Data Samples And Volume:

Sample Files:

We will collect the reports from the SBP site and will provide the sample raw files which will represent both full load and incremental load type of data collection.

The full load sample will be the initial load/collection of the data and will represent the historical reports whereas the incremental load sample will represent the newer quarterly report added after the historical data.

Expected Volume and Frequency:

The reports are mainly pdf style documents which contains tables so the raw data volume is expected to be relatively small in comparison with the large scale commercial datasets and since we have a sample raw file which has memory of 1.1 mb so we know that each year has 4 quarters so for each year we would have around 4.4mb and for full load we would have 4.4×20 which would be equal to 88.8mb.

We expect the complete historical size of the initial load(Full Load ingestion) would be in tens to hundreds of MB, depending on the number and size of reports we collect and every year one new report would be added to the already ingested data.

3- Security and Compliance:

The SBP payment systems review contains the aggregated banking as well as the payment stats rather than the simple individual customer record(if they stored every customer individual transaction record then the data would not be something with which we can easily work with).

Our expected data includes the information explained below:

- Number of transactions
- Transaction values
- Number of ATMs
- Number of POS terminals
- Number of payment cards
- Number of digital banking users
- Number of merchants

Since we already know that the dataset does not contain the individual information of any customer so we do not expect that the dataset would have any personally identifiable information. And since the data is published as aggregated stats by the SBP, so hashing or masking of individual customer information will not be required but we would still review the raw reports before processing the information as to be sure that no confidential or sensitive information is included in the analysis.

4- High Level Medallion Data Modelling:

The bronze layer is the most initial layer in the architecture and will contain all the original data which is collected from the SBP site without any changes. The purpose of this layer is to preserve the original information before cleaning.

The raw reports will be stored along with information such as:

- Report year
- Quarter
- Source report
- Data collection date

Silver Layer:

In the following silver layer, the information would be cleaned and converted into a format which would be consistent. Some expected processing includes:

- Removing unnecessary text from extracted tables
- Standardizing column names
- Converting transaction values into numeric values
- Converting dates and quarters into a standard format
- Handling missing values
- Removing duplicate records
- Making units consistent across different reports
- Standardizing payment channel names

The silver layer data could be organized in such ways: payment-transaction, digital-banking, payment cards, payment infrastructure, ecommerce.

Gold Layer:

The gold layer would contain summarized data that can be directly used for analysis and for creating and using for dashboards.

Possible Tables are:

Payment-trends, digital payment summary, payment channel summary, infrastructure summary, ecommerce summary.

We will use the gold layer to make it easier to answer business and analytical questions without repeatedly processing the raw data and using the data already pre processed from the silver layer.

These tables could contain information such as:

- Quarterly transaction volume
- Quarterly transaction value
- Growth in digital payments
- Payment channel comparisons
- Growth in ATMs and POS terminals
- Growth in digital banking users
- E-commerce transaction trends

5- Business Intelligence and Dashboard:

The final dashboard which we would be creating would be using the power bi software. The purpose why would we be creating the dashboard is: to show how the Pakistan's payment system has changed from traditional payment methods such as using only cash and moving towards digital payment channels.

The business questions which the dashboard will answer will be following:

- Which payment channels have experienced the largest growth?
- How has the share of digital payments compared with the traditional only cash payments changed over time?
- How has the average transaction size changed across time?
- How have the transaction volume and transaction value changed over time?

Visual Charts:

- 1- Line chart: it would show transaction value and transaction volume over the years for different payment channels.
- 2- KPI cards: which shows total transactions, total transaction value, number of payment channels, etc.
- 3- Bar charts: we can use it to show the growth of the payment channel.

6- Infrastructure and Cost Management:

Github repo: <https://github.com/noyansiddiqui99-maker/Pakistan-s-Digital-Payment-Analytics>

We are planning to use Azure DataBricks (free edition) for processing the data with Apache Spark and power bi software for dashboard creation. Since the SBP reports are relatively smaller we wont need much computing resources and the cost would be very small.

Reason:

We will first store the original data, then process the data and then test the pipeline on a sample before we run the pipeline on the report's information, then we will use the initial load and process only new available data and would avoid the data to be added which is already present before. These above are the given reasons for why we would be able to complete the project within the free edition limits of azure databricks.